



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 3 .08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No	: BIO G524
Course Title	: Animal Cell Technology
Instructor-in-Charge	: SUDESHNA MUKHERJEE
Instructors & Practical Instructors	: Soumitra Ghosh, Satya Om Prabha & Saswata Chatterjee

1. Course Description:

Animal cell and tissue culture techniques from various organisms, basics of primary cell culture, types of cell lines, generation and maintenance of cell lines, current advanced technology for cell biology application, manipulation and applications of cell culture technology for Biotechnological research and its therapeutic application.

2. Scope and Objective of the Course:

In this course we will discuss:

- Biology of cultured cells & cell microenvironment
- Maintenance, culture cryopreservation and Quantitation of cells
- Primary cultures and cell lines
- Cytotoxicity, proliferation and Cell Transformation
- Cellular Imaging Techniques and its application
- Scaling up and Automation.

3. Text Books:

TB 1. Freshney, R.I. Culture of Animal Cells: A Manual of Basic Technique and Specialized Applications, Willey-Blackwell Press (6th Ed), 2010.

4. Reference Books:

1. **R1:** Asok Mukhopadhyay. Animal Cell Technology. I. K. International Publishing House Pvt. Ltd. 2009.
2. **R2:** Robert A. Weinberg. *The Biology of Cancer*., Garland Science. 2014.

Note: Recent review articles/Papers/research covering advanced cell biology techniques and its application will be discussed and corresponding material will be provided to you time to time.

5. Course plan:



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Module Number	Lecture Session	Reference	Learning Objective
1. Introduction	L1.1.Types of culture	TB1 Chapter 1	Basic exposure to Types, application and advantages of tissue culture
	L1.2. Advantages and limitations of tissue culture		
2. Biology of cultured cells	L2.1 The Culture microenvironment.	TB1Chaper 2	To learn about the behavior of cultured cells at a molecular level during its growth, proliferation and differentiation.
	L2.2 Cell Adhesion, Proliferation and Differentiation		
	L2.3 Cell Signaling		
3.Bio safety and lab ethics	L3.1 Laboratory safety and ethics.	TB 1 chap 6 & RB1 Chapter 4)	To know the practical aspects of Biosafety and understand the significance of containment levels
	L3.2 Levels of Biological Containment		
4. Preparations for Culturing Cells - aseptically	L4.1 Aseptic techniques, sterilization.	TB 1 chap 4, 5 & RB1 Chapter 6	Learning the important practical perquisites and procedures to grow cells
	L4.2 Defined media and supplements, serum-free medium.		
5. Primary cultures, sub cultures and cell lines	L5.1 Types of primary cell cultures, isolation of tissue, primary culture.	TB 1 chap 11, 12, 13, 14 & RB1 chap 9	Learning to grow cells from tissue/ organ and quantitative analysis
	L5.2 Quantitation of cells		
6. Cell characterization and transformation	L. 6.1 Characterizing cells in the culture.	TB 15 chap 6 &	Learning the important parameters of characterizing cells.
	L6.2 Transformation,		
	L6.3 Immortalization & Tumorigenicity		
7.Contamination	L7.1 Source & Types of contamination,.	TB chap18	Learning how to recognize, avoid and eliminate contamination
	L7.2 Monitoring and eradication of contamination		
8.Cytotoxicity	L8.1 Viability, Toxicity & Survival	TB chap 21	Understanding the measurement and application of cytotoxic drugs
	L8.2 Application of Cytotoxicity Assays		
9. Scale- up and Automation	L9.1 Scaling up of suspension culture	TH Chap 26	To learn how to transfer lab scale culture work to an industrial scale
	L9.2 Scaling up of monolayer cultures		



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	L9.3 Process control and Automation		
10. Specialized techniques and implications of cell culture	L10.1 Viable cell separation & Confocal microscopy	Chap. 26	To understand the concept and working of specialized techniques used in animal cell technology
11. Therapeutics application of cell culture	L11.1 Cloning & Gene Therapy	RB1 Chap 14 & 15	To understand how animal cell technology can be used to treat diseases
	L11.2 Regenerative medicine		
12. Genetic manipulation of experimental animals	L12.1 Experimental animal models	Review and Research article	To see how genetically manipulated animals are being used in medicine and research.
	L12.2 Somatic cell fusion & monoclonal antibodies		
13. Advanced cell Based Assay Models	L13.1 Learning about several assays that been currently employed for cellular functional studies.	Review and Research article	To explore cell biology tools and models to understand cellular structure and functions.

Lab Plan: Hands On Experience will be provided on the Following Aspects

- Sterilization Techniques
- Preparation of Animal Cell Culture related media, antibiotic and buffer solutions
- Freezing and revival of Cell lines
- Maintenance of Adherent/Non adherent Cell Lines
- Cell Counting
- Staining and viability testing of cells
- MTT Assay/Cytotoxicity Assay for screening
- Fluorescence Cell Imaging (DAPI staining)

***Completion of all wet lab experiments are dependent on availability of chemicals and resources.**

6. Evaluation scheme:

Component	Duration	Weightage (%)	Date and time	Nature of component Close Book (CB)/ Open Book (OB))
Mid Semester test	90 mins	20%		CB/OB
Surprise and announced Quizzes	15min	10%		CB



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Experiments and Lab quiz		30%		CB & OB
Seminars/ Assignment		10%		OB
Comprehensive	2 hrs	30%		CB & OB

7. Chamber Consultation Hour: Chamber consultation hour will be announced in the class.

8. Notices: The notices will be displayed in Google Class Room/Nalanda.

9. Make-up Policy: Make-up will be granted for genuine cases only. Certificate from authenticated doctor from the Medical Center must accompany make-up application (only prescription or vouchers for medicines will not be sufficient). Prior permission of IC is compulsory. No make up for tutorial tests.

10. Note (if any):

10.1 Academic Conduct Policy: It is expected that all students follow the highest standard of academic practice when participating in any evaluation component. Having a zero-tolerance for academic dishonesty, any case of misconduct, however minor, will be dealt with appropriately. The case may be reported to the Dean, Instruction Division, if necessary.

10.2 Grading Policy: Award of grades would be guided by the histogram of marks and course average. If a student absents himself/herself in any component (listed in the Evaluation Scheme), his/her performance in the course may be reported as 'NC' (Not Cleared).

10.3 Policy for Not cleared grade (NC): Any student getting less than 40% of median and 30% of the average of top 3 students (whichever is low) is liable to get a NC Grade.

Instructor-in-charge
Course No. BIO G524